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Determinants of serum half-lives for linear and branched perfluoroalkyl substances after long-term high exposure—A study in Ronneby, Sweden

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ABSTRACT

Background: Per- and polyfluoroalkyl substances (PFAS) are persistent substances with surfactant and repellent properties. Municipal drinking water contaminated with PFAS had been distributed for decades to one third of households in Ronneby, Sweden. The source was firefighting foam used in a nearby airfield since the mid-1980s. Clean water was provided from December 16, 2013.

Aims: The purpose was to estimate serum half-lives and their determinants in the study population for different PFAS.

Methods: Up to ten blood samples were collected between 2014 and 2018 from 114 participants (age 4–84 years at entry, 53% female). 19 PFAS were analysed. Linear mixed models were used to estimate the half-lives.

Results: Eight PFAS were increased in Ronneby: perfluorooctanoic acid (PFOA), perfluoropentane sulfonate (PFPeS), perfluorohexane sulfonate (PFHxS), perfluoroheptane sulfonate (PFHpS), linear perfluorooctane sulfonate (L-PFOS) and three branched perfluorooctane sulfonates (1 m-PFOS, 3/4/5m-PFOS and 2/6m-PFOS). The mean estimated half-lives (in years) were 0.94 (95 %CI 0.86–1.02) for PFPeS, 2.47 (2.27–2.7) for PFOA, 2.67 (2.51–2.85) for 2/6m-PFOS, 2.73 (2.55–2.92) for L-PFOS, 3.43 (3.19–3.71) for 3/4/5m-PFOS, 4.52 (4.14–4.99) for PFHxS, 4.55 (4.14–5.06) for PFHpS, and 5.01 (4.56–5.55) for 1 m-PFOS.

The most important determinants of a shorter half-life were young age, and better kidney function measured by estimated glomerular filtration rate and ratio of paired urine and serum PFAS levels, followed by female sex during their fertile period aged 15–50. Markers of gut inflammation and reduced permeability i.e. zonulin and calprotectin were also possibly associated with shorter half-life. The results also suggested a time-dependent PFAS elimination process, with more rapid elimination in the first year after the end of exposure.

Conclusion: The half-life estimates are in line with past estimates for some PFAS such as PFOA, and the novel results for different PFOS isomers. These results provide observational support for elimination routes – renal, fecal and maternal.

1. Introduction

Per- and polyfluoroalkyl substances (PFAS) are persistent substances with surfactant and repellent properties. The production and use of PFAS has led to global environmental contamination. The source of exposure to PFAS in the general population may include food, house dust, drinking water, and, for infants, placental transfer and breast milk (Begley et al., 2005; Berger et al., 2009; Falandysz et al., 2006; Shoeib et al., 2005). Locally, drinking water contamination is a significant exposure source, particularly at very high exposure levels identified in four large communities in the mid-Ohio Valley US (C8 population); Arnsberg, Germany; Veneto, Italy cohort and in Ronneby, Sweden (Brede et al., 2010; Frisbee et al., 2009; Pitter et al., 2020; Xu et al., 2021) but also at lower general population levels (Glynn et al., 2020; Hu et al., 2019). A wide range of health effects of PFAS during development, as well as later in life, are suspected even at present background human

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exposure (EFSA, 2018).

It is imperative to establish the rate of elimination of PFAS from the body after cessation of exposure, not only for modelling purpose in epidemiological studies, but also for risk assessment and risk communication in affected areas worldwide. Various animal studies have estimated serum half-lives ($t_{1/2}$) for PFAS in different species, and reported half-lives usually in hours or days with markedly variation between species (Lau et al., 2007). Human half-lives for the few PFAS studied are substantially longer than animals (Bartell et al., 2010; Li et al., 2018; Olsen et al., 2007; Xu et al., 2020a). Half-lives in human have been reported for perfluorooctane sulfonic acid (PFOS) and perfluorooctanoic acid (PFOA) for between 2 and 4 years, and perfluorohexane sulfonic acid (PFHxS) for 5 up to 7, while some short-chain PFAS have considerably shorter half-lives (Xu et al., 2020a). There are some differences in the average half-life reported between different human studies, and in addition we have demonstrated there is substantial inter-individual variation for which there is very little explanation (Li et al., 2018).

However, these studies have provided very few insights into the determinants of PFAS elimination. For example, it is expected that females in fertile ages have a shorter PFAS half-life than males due to by menstrual blood losses, pregnancies and breast-feeding (Wong et al., 2014). For PFOA, dose-or time-dependent elimination has been suggested (Brede et al., 2010; Seals et al., 2011; Xu et al., 2020a). In a C8-study, the elimination rate was related to single nucleotide polymorphisms (SNPs) in tubular transporter proteins (Fletcher et al., 2012), indicating that renal elimination phenotypes may be important, although a smaller cross sectional analysis of PFOA levels in relation to one of these SNPs in an Italian study did not find any significant association (Ingelido et al., 2020). Therefore there remains a need for more detailed information on half-life for more PFAS than the few most-studied and the variation by age, sex, body mass index and other potential determinants. Given that elimination can be by both renal and gut excretion, factors that affect these routes are of interest to explore whether they may be reflected in individual PFAS half-life.

An earlier paper on this same population derived half-lives for PFHxS, PFOA and PFOS (Li et al., 2018), and the current paper provides improved estimates for half-lives in this population with longer follow-up and an extended analysis of other PFAS compounds.

The aims in this paper were to estimate serum half-lives of a range of perfluoroalkyl substances with varying carbon chain-length after the abrupt cessation of long-term high exposure to contaminated drinking water, and to characterize the individual determinants of serum half-lives.

2. Material and method

2.1. Study group

In 2013, it was found out that drinking water from one of two municipal waterworks in Ronneby, a municipality with 28 000 inhabitants in southern Sweden, was contaminated with very high levels of PFAS (Xu et al., 2021). This waterworks, which served approximately one third of the households in the municipality, was closed on 16 Dec 2013 and water was immediately provided from the other non-contaminated waterworks. It was soon confirmed that use of aqueous film-forming foam (AFFF) for firefighting at the nearby military airport since the mid-1980s was the source of contamination.

Extensive biomonitoring in the municipality started in June 2014, approximately 6 months after end of the exposure, by open invitation and free of cost. Subjects for a half-life study were recruited among the first 1000 participants, until the target of 100 subjects was reached with a wide range of serum levels and ages. These participants have donated so far serum samples in 10 rounds: in June 2014, October 2014, January 2015, April 2015, September 2015, March 2016, September 2016, April 2017, September 2017 and May 2018. Urine samples were collected in September 2015 (round 5), March 2016 (round 6) and May 2018 (round 10). Fecal samples were collected in September 2015. Ongoing sampling is planned for several years to come.

A reference group from Karlshamn, a nearby municipality without PFAS contamination in drinking water, was also examined in 2016 (Li et al., 2018; Xu et al., 2021). 68 of the participants in the reference group, with age range 20–50 years, were randomly selected to estimate the background exposure level for an extended range of PFAS compounds.

2.2. Chemical analysis

2.2.1. PFAS in serum and urine

Venous serum samples were obtained, left at room temperature for 30 min and then centrifuged for 10 min at 1500g to isolate the serum on site. The samples were collected in 5 ml Becton Dickinson vacutainer blood collection tubes without gel (BD, Belliver Industrial Estate, Plymouth, UK). Serum samples were stored in 2 ml micro tubes (Sarstedt, Nümbrecht, Germany). Urine samples were collected in 13 ml polypropylene screw cap tubes (Sarstedt, Nümbrecht, Germany). Serum samples were kept on dry ice and urine samples were maintained cool by ice blocks after sampling and transported to Lund using cold chain logistics. Serum and urine samples were stored at −80 °C and −20 °C, respectively, before analysis.

Quantitative analysis of PFAS was conducted at the Division of Occupational and Environmental Medicine, Lund University, using liquid chromatography - triple quadrupole linear ion trap mass spectrometry (LC/MS/MS; QTRAP 6500+; AB Sciex, Framingham, MA, USA) according to Xu 2020a. In total, 19 PFAS compounds were quantitated in the panel group: perfluoropentanoic acid (PFPeA), perfluorohexanoic acid (PFHxA), perfluoroheptanoic acid (PFHpA), perfluorooctanoic acid (PFOA), perfluorononanoic acid (PFNA), perfluorodecanoic acid (PFDA), perfluoroundecanoic acid (PFUnDA), perfluorododecanoic acid (PFDoDA), perfluorotridecanoic acid (PFTrDA), perfluorobutane sulfonic acid (PFBS), perfluoropentane sulfonic acid (PFPeS), perfluorohexane sulfonic acid (PFHxS), perfluoroheptane sulfonic acid (PFHpS), linear perfluorooctane sulfonic acid (L-PFOS), perfluoro-1-methylheptanesulfonate(1m-PFOS), perfluoro-3/4/5-methylheptane sulfonate (3/4/5m-PFOS), perfluoro-2/6-methylheptanesulfonate (2/6m-PFOS), perfluorononane sulfonic acid (PFNS) and perfluorodecane sulfonic acid (PFDS). 2/6m-PFOS is the sum of 2 m-PFOS and 6 m-PFOS, since the estimating method could not distinguish them from one another. Likewise, 3/4/5m-PFOS is the sum of 3 m-PFOS, 4 m-PFOS and 5 m-PFOS. Briefly, 25 μL of serum or 50 μL of urine samples were precipitated using acetonitrile by vigorous shaking for 30 min prior to analysis. All sample batches include chemical blanks and 3 quality control samples. The analyses of PFOS and PFOA are part of a quality control program between analytical laboratories coordinated by Professor Hans Drexler, Institute and Outpatient Clinic for Occupational, Social and Environmental Medicine, University of Erlangen-Nuremberg, Germany. The laboratory also participated in the HBM4EU QA/QC program, and its successful performance has resulted in its qualification as HBM4EU laboratory for the analysis of: PFPeA, PFHpA, PFOA, PFNA, PFDA, PFUnDA, PFDoDA, PFBS, PFHxS, PFHpS and PFOS. The limit of detection (LOD) in the serum and urine samples are listed in Table S1. Density was used to compensate for urine sample dilution and was measured using a digital refractometer (Hauser et al., 2004). For further details see Chemical analysis in (Xu et al., 2021) and Supplementary materials.

2.2.2. Biomarkers

Cystatin C were analysed in serum at the Department of Clinical Chemistry of the Skåne University Hospital in Lund, Sweden, on a Cobas 8000 analyzer (Roche Diagnostics, Basel, Switzerland). The laboratory is accredited according to ISO15189.

Calprotectin and zonulin were analysed in fecal samples, which were stored at − 80 °C and analyzed within a few months after collection. The

calprotectin concentration was determined by an enzyme-linked immunosorbent assay (ELISA) kit (Immundiagnostik AG, Bensheim, Germany, Batch number K6927-161 124) based on the two site sandwich technique The zonulin concentration was determined by an ELISA kit (Immundiagnostik AG, Batch number K5600-160 519) using the competitive binding technique. For further description, see (Xu et al., 2020b).

2.3. Determinants

2.3.1. Age group

The participants were divided into three age groups i.e. ≤ 14, >14–50, >50, at first sampling, broadly reflecting pre-and early puberty, menstruating ages and postmenopausal ages.

2.3.2. Body mass index (BMI)

For those above 14 years, we have investigated the possible BMI effect on elimination. BMI was dichotomized at 25, i.e. normal weight vs overweight. There was no one in the study group above 14 years that was underweight i.e. BMI below 18.5.

2.3.3. Kidney and gastrointestinal function

For the kidney, the glomerular filtration rate is a marker of kidney function, and for the gut, calprotectin and zonulin are markers of inflammation and permeability, which may affect PFAS reabsorption.

Estimated glomerular filtration rate (eGFR) was used as indicator of kidney function and based on equations using serum cystatin C based on CAPA formula (Grubb et al., 2014). eGFR was divided into two categories above or below 90 ml/min/1.73 m^2.

For those having donated fecal samples (n = 57), measured zonulin and calprotectin levels were used as indicators of gastrointestinal permeability and inflammation (Walsham and Sherwood, 2016; Malíčková et al., 2017). Zonulin was dichotomized at their median levels. A conventional cut-off at 50 mg/kg was used for calprotectin (Konikoff and Denson, 2006).

2.3.4. Urine/serum ratio

62 participants had donated urine samples in rounds 5, 6 and round 10. All of the 8 PFAS at higher concentrations with most above 1 ng/ml (PFOA, PFPeS, PFHxS, PFHpS, L-PFOS, 1 m-PFOS, 3/4/5m-PFOS, 2/6m-PFOS) were detected in the most of urine samples. Based on the paired urine and serum concentration, their ratios were calculated. The denominator is the serum concentration, a surrogate of the body burden of PFAS. The numerator is the urine concentration, reflecting the renal excretion pathway. Based on the ratios, each participant could be classified as having high or low renal excretion depending on the ratio's relation to the median of the specific round. Since there were up to three rounds of urine samples, a participant who consistently remained in the high ratio category in all the three rounds urine samples was classified as having a 'high' ratio. We used the urine values measured even if below the LOD. These were used in the numerator for the urine/serum ratio and then they were classified into high vs low ratios. Thus those below LOD were rightly placed in the low category.

2.4. Statistical analysis

Serum PFAS half-lives were estimated based on linear mixed model as in Li et al (2018). The following mixed model was used to fit the panel data:

$$\ln C_{ij} = \alpha_i + t_{ij}k_i + X_i\beta + \varepsilon_{ij} \quad (1)$$

where C_{ij} is the serum PFAS concentrations for individual i and sampling round j after subtracting the geometric mean of the background levels. If the serum PFAS levels after subtraction were lower than the half of the background levels, they were replaced with half of the background levels. In all, 26 measures of PFOA from seven participants and 10 measures of serum PFPeS from six participants were replaced in this way. α_i is the subject-specific intercept, t_{ij} is the time elapsed between the end of distribution of contaminated water and the blood sample collection, k_i is the subject-specific slope, X_i is a vector of fixed covariates for individual i, including age, gender and BMI, β is the fixed effect coefficient and ε_{ij} is the random error term. For one child BMI at baseline was not reported, and thus the BMI was estimated by the average of the participants in the same age. The subject-specific intercept α_i, the subject-specific slope k_i and the random error term ε_{ij} were modeled as random with normal distribution; others were treated as fixed effects. The slope k_i is the individual elimination rate and its mean estimated from model (i.e. population average elimination rate) was converted to half-life (ln2/mean(k_i)) to derive the mean half-life. Individual values of k_i were predicted using the "best linear unbiased prediction" method, i.e. an individual elimination rate for each participant derived from the model via command 'random /solution' in the procedure 'proc mixed' in SAS. Summary half-life values have been presented as either a mean half-life (calculated from the mean elimination rate constant k: $t_{1/2} = \ln(2)/k$) or as median half-life (the median value of the individual predicted half-life values). The 95% CI for mean (k_i) from the regression was used to derive the CI for the half-life, by converting as for the mean.

We conducted additional analyses to determine whether elimination rates depended on certain covariates. In addition to model 1, we added interaction terms between age group (≤14, >14–50, >50) and elapsed time, interactions between age group, gender and elapsed time, and the interactions between initial total PFAS levels tertiles and elapsed time.

Since age group had a strong effect on elimination, the interaction term between elapsed time and age groups were kept in following analysis. For a subgroup with age over 14, we adding interaction of dichotomous BMI variable (below or above 25) with elapsed time.

We also evaluated the potential dependence of elimination rates on time elapsed after end of exposure by comparing the estimates based on the samples from round 1 to round 3, (6–13 months after end of drinking water exposure), round 3 to round 6 (13–25 months) and round 6 to round 10 (25 to 53 months).

In addition, we evaluated whether kidney function measured by eGFR, gut integrity measured by zonulin and calprotectin, and urine excretion measured by urine/serum ratio have an influence on elimination rate by adding interaction terms between these dichotomous variables and elapsed time.

Finally, we calculated partial R^2, as a measure of the relative importance of the determinants based on the linear model with the individual predicted half-life as outcome and the determinants as covariates i.e. interaction terms between age and sex, categorical eGFR, zonulin, calprotectin and initial total PFAS levels, all in the model together.

2.5. Ethics

The study has been carried out in accordance with the Declaration of Helsinki. The study was approved by the Regional Ethical Board in Lund, Sweden. All participants have given informed consent prior to study participation.

3. Results

3.1. Study participants.

The panel study group consisted of 114 participants (53% females) with a wide age span, from 4 to 84 years at baseline (Table 1). The average number of serum samples was 8. There were 29 children with age below 14, 52 participants aged 14–50 and 33 over age 50, at baseline. Out of those over age 14, there were 48% having BMI over 25. 109 participants had valid Cystatin C levels. There were 53 participants

Table 1
Description for the 114 participants in Ronneby panel study.

	Counts	Median (5,95 percentiles)	Counts in each category
Age (years)	114	42 (7, 76)	<14: 29; 14–50: 52; >50: 33
BMI (kg/ m^2)	114	24 (16, 32)	>18.5: 17; 18.5–25: 55, >25 42 for age > 14: BMI < 25: 43; >25: 40.
eGFR (ml/min/1.73 m^2)	109	91 (47, 119)	≥90:47; 60–90:53; <60: 9
Zonulin (ng/ml)	57	106 (20, 323)	≥106: 29; <106: 28
Calprotectin (mg/kg)	57	20 (2, 135)	>50: 13; <50: 44

having eGFR < 90 ml/min/1.73 m^2 and 9 participants with eGFR < 60 ml/min/1.73 m^2.

3.2. Serum PFAS levels

In total 19 PFAS compounds were measured in the panel group (Table 2). At 1st sampling, PFHxS showed the highest serum concentration, with geometric mean level 480 times higher than the level observed in the reference population. PFOA and PFHpS also showed substantial higher levels. The levels of PFPeS>100 times higher than in the reference. The linear and branched PFOS substances (L-PFOS, 1 m-PFOS, 2/6m-PFOS and 3/4/5m-PFOS) were substantially higher with 50–150 times higher levels in the panel group than in the reference group. We focused on these 8 predominant PFAS compounds in the following analyses.

3.3. Urinary PFAS levels

Compared to serum PFAS levels, the urinary PFAS levels were low, with a median for PFHxS 0.13 ng/ml and for other PFAS medians were below 0.08 ng/ml (Table 3). At the group level, the urine/serum ratios for PFPeS were higher than the rest of PFAS compounds.

For each PFAS, there was a substantial between-individual variation of the urine/serum ratio. The ratios were more consistent over time as the median of ratios were similar across times. Over 60% of the participants showed consistency in high or low urine/serum ratio over the first two rounds, i.e. a participant had relatively high urine/serum ratio at sampling round 5 had also high ratio at the sampling round 6. Around 40–60% participants showed consistently high or low ratios in all three rounds (Table S2).

There were indications of associations of a decreased urine/serum ratio with increased serum PFAS levels (the scatter plots are shown in supplementary Figure S1). There were no associations between urine/serum ratio and eGFR (Supplementary Figure S2).

3.4. Half-lives of PFAS

From June 2014 to May 2018, a four-year period, the geometric mean serum level decreased 56% PFOA, 92% for PFPeS, 43% for PFHxS, 38% for PFHpS, 60% for linear PFOS (L-PFOS), 40% for branched 1 m-PFOS, 54% for branched 3/4/5m-PFOS and 64% for branched 2/6m-PFOS.

Table 4 shows the elimination constant and the corresponding half-lives for each PFAS compound. PFPeS showed the most rapid annual decrease with 52%, which corresponds to a mean half-life of 0.94 years. PFOA, L-PFOS and 2/6m-PFOS had half-lives around 2.5 to 3 years. 3/4/5m-PFOS showed a half-life of 3.4 years. PFHxS, PFHpS and 1 m-PFOS showed relative longer half-lives than rest with 4.5 years, 4.6 years and 5.0 years respectively.

The sensitivity analysis with serum PFAS levels not subtracting general background levels are listed Table S3. The estimates without subtraction of background levels were very close to estimates after subtraction of background exposure, except to an extent for PFOA (2.99 vs 2.47 years).

Table 2
Serum PFAS levels (ng/ml) in panel group and reference group.

	Panel Study group (n = 114)								Reference group (n = 68)			
	Initial PFAS levels (0.5 years)				PFAS levels at 10th round (4.4 years)							
	GM	Median	min	max	GM	Median	min	max	GM	Median	min	max
PFPeA	<0.08	<0.08	<0.08	0.090	<0.08	<0.08	<0.08	<0.08	<0.2	<0.2	<0.2	<0.2
PFHxA	0.099	0.094	<0.07	1.2	0.074	0.082	<0.07	0.65	<0.2	<0.2	<0.2	<0.2
PFHpA	0.085	0.095	<0.05	2.4	<0.05	<0.05	<0.05	0.11	<0.1	<0.1	<0.1	<0.1
PFOA	16	16	1.7	78	7.1	6.7	0.87	46	1.4	1.4	0.26	4.5
PFNA	0.64	0.63	0.19	3.9	0.51	0.52	0.12	1.3	0.63	0.64	0.22	1.4
PFDA	0.27	0.26	0.11	1.6	0.24	0.24	0.058	0.67	0.27	0.27	0.11	0.76
PFUnDA	0.27	0.27	<0.1	1.3	0.23	0.26	<0.1	0.68	0.23	0.21	0.098	0.66
PFDoDA	<0.2	<0.2	<0.2	0.29	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2
PFTrDA	<0.2	<0.2	<0.2	0.33	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2	<0.2
PFBS	<0.05	<0.05	<0.05	1.6	<0.05	<0.05	<0.05	0.10	<0.1	<0.1	<0.1	<0.1
PFPeS	6.2	6.0	0.083	49	0.48	0.51	<0.02	9.8	<0.02	<0.02	<0.02	0.018
PFHxS	260	260	15	1300	150	140	12	860	0.54	0.61	<0.1	1.2
PFHpS	13	12	0.91	70	8.0	8.0	0.55	51	0.059	0.069	<0.1	0.16
L-PFOS	150	170	12	680	63	67	10	500	3.0	3.0	0.30	9.8
1 m-PFOS	23	23	1.7	110	14	13	1.0	86	0.17	0.18	<0.05	0.59
3/4/5m-PFOS	73	75	6.4	410	34	34	3.5	220	0.76	0.84	0.096	2.2
2/6m-PFOS	49	56	4.3	270	17	20	2.7	130	0.47	0.47	<0.3	1.4
PFNS	<0.05	<0.05	<0.05	0.055	<0.05	<0.05	<0.05	<0.05	<0.1	<0.1	<0.1	<0.1
PFDS	<0.1	<0.1	<0.1	0.23	<0.1	<0.1	<0.1	<0.1	<0.1	<0.1	<0.1	<0.1

Note: Perfluoropentanoic acid (PFPeA), Perfluorohexanoic acid (PFHxA), Perfluoroheptanoic acid (PFHpA), Perfluorooctanoic acid (PFOA), Perfluorononanoic acid (PFNA), Perfluorodecanoic acid (PFDA), perfluoroundecanoic acid (PFUnDA), perfluorododecanoic acid (PFDoDA), perfluorotridecanoic acid (PFTrDA), Perfluorobutane sulfonic acid (PFBS), Perfluoropentane sulfonic acid (PFPeS), Perfluorohexane sulfonic acid (PFHxS), Perfluoroheptane sulfonic acid (PFHpS), Linear perfluorooctane sulfonic acid (L-PFOS), Perfluorononane sulfonic acid (PFNS), Perfluorodecane sulfonic acid (PFDS), Perfluoro-1-methylheptanesulfonate (1 m-PFOS), Perfluoro-3/4/5-methylheptanesulfonate (3/4/5m-PFOS), Perfluoro-2/6-methylheptanesulfonate (2/6m-PFOS).

Table 3
Density adjusted PFAS urinary levels (unit: ng/ml).

		Density adjusted urine levels			Serum levels	Urine/serum ratio
	Round	median	min	max	median	median
PFOA	5[a]	0.025	<0.01	0.16	12	0.0023
	6[b]	0.025	<0.01	0.12	10	0.0021
	10[c]	0.017	<0.01	0.092	6.7	0.0022
PFPeS	5	0.028	<0.01	0.18	2.6	0.0098
	6	0.030	<0.01	0.75	1.9	0.0030
	10	0.012	<0.01	0.078	0.51	0.017
PFHxS	5	0.18	0.016	1.4	220	0.00072
	6	0.17	0.019	0.64	210	0.00075
	10	0.13	0.014	1.9	140	0.00075
PFHpS	5	<0.01	<0.01	0.071	11	0.00086
	6	0.011	<0.01	0.036	10	0.00099
	10	<0.01	<0.01	0.057	8.0	0.00085
L-PFOS	5	0.073	<0.01	1.4	130	0.00070
	6	0.075	<0.01	0.74	120	0.00068
	10	0.050	<0.01	0.92	67	0.00073
1 m-PFOS	5	<0.01	<0.01	0.062	20	0.00024
	6	<0.01	<0.01	0.048	19	0.00021
	10	<0.01	<0.01	0.14	13	0.00021
3/4/5m-PFOS	5	0.039	<0.02	0.42	60	0.00073
	6	0.035	<0.02	0.25	59	0.00067
	10	0.031	<0.02	0.48	34	0.00086
2/6m-PFOS	5	<0.02	<0.02	0.23	42	0.00041
	6	<0.02	<0.02	0.11	38	0.00045
	10	<0.02	<0.02	0.15	19	0.00052

Note: Perfluorooctanoic acid (PFOA), Perfluoropentane sulfonic acid (PFPeS), Perfluorohexane sulfonic acid (PFHxS), Perfluoroheptane sulfonic acid (PFHpS), Linear perfluorooctane sulfonic acid (L-PFOS), Perfluoro-1-methylheptanesulfonate (1 m-PFOS), Perfluoro-3/4/5-methylheptanesulfonate (3/4/5m-PFOS), Perfluoro-2/6-methylheptanesulfonate (2/6m-PFOS). [a] Round 5: twenty-one months after end of exposure with 104 samples. [b] Round 6: twenty-seven months after end of exposure with 93 samples. [c]Round 10: four and half years after end of exposure with 73 samples.

There was a substantial inter-individual variation of the estimated half-lives with the three to five fold difference from the 5th to 95th percentiles (Table 4).

Half-life could not be estimated for the other 11 PFAS with at least some measures above detection, for various reasons, including that they were nearly all below detection (e.g. PFPeA), or not present in the contaminated water and so exposure must come from other routes such as diet (e.g. PFDA) or they were in the contaminated water but excretion is so rapid that nearly all samples were below detection by the time of sampling the first round (e.g. PFBS).

3.5. Determinants of half-lives of PFAS

We explored the possibility of a change in elimination rate over time since end of exposure (a combination of time- and dose-dependency). The elimination rates based on the samples during the first year of observation after end of exposure were generally faster than the elimination based on the late samples, indicating the possibility of time-dependency (Table 5). A time-dependency of elimination was not obvious for the branched PFOS, i.e. 1 m-PFOS, 3/4/5m-PFOS and 2/6m-PFOS.

Initial PFAS levels were relevant as well, with positive associations between higher initial PFAS levels and shorter half-lives in some of the PFAS substances (Fig. 1, Table S4), especially in PFHxS and PFHpS.

Age had a very strong effect on half-lives especially for the preteens, as shown in Fig. 2 (details in Table S5). In general, shorter half-lives were associated with younger ages. The preteens showed 45%-60% shorter half-lives for different PFAS, comparing with those older than 50, for all PFAS except PFPeS which showed the most rapid excretion in the mid age range. There was also some evidence for the shorter half-lives in adults aged below 50 than aged above 50. The scatter plot of individual elimination rate vs age suggest a fairly steady decline in excretion and consequent increase in half-life throughout the lifetime (Supplementary Figure S3).

An effect of gender on half-life was only evident between 15 and 50 years with female having shorter half-lives (5–22%) for all PFAS compounds expect PFPeS (Fig. 3, details Table S6-8). There was no evidence of BMI having an effect on half-lives in adults (Table S9).

It is evident that the participants with lower eGFR have 12–27% longer half-lives for all the PFAS compounds (Fig. 4, Table S10).

The participants with higher urine to serum ratio had short half-lives for all the PFAS (Fig. 5, Table S11).

Fecal calprotectin levels (Fig. 6, Table S12) showed negative associations with half-lives, although the difference only reached statistical significance for PFOA ($p = 0.02$). Participants with calprotectin > 50

Table 4
Population average constant elimination rate (per year) and half-life (in years) for each PFAS, n = 114.

	Mean Elimination Rate [a] (95% CI)	Percent annual decrease	Mean half-life[b] (95% CI)	Median half-life (5th, 95th percentiles) [c]
PFOA	−0.28(-0.31, −0.26)	24.50%	2.47(2.27, 2.7)	2.69(1.37, 5.4)
PFPeS	−0.74(-0.8, −0.68)	52.35%	0.94(0.86, 1.02)	1.03(0.47, 1.81)
PFHxS	−0.15(-0.17, −0.14)	14.20%	4.52(4.14, 4.99)	5.4(2.34, 9.29)
PFHpS	−0.15(-0.17, −0.14)	14.12%	4.55(4.14, 5.06)	5(2.3, 11.95)
L-PFOS	−0.25(-0.27, −0.24)	22.45%	2.73(2.55, 2.92)	2.89(1.62, 4.75)
1 m-PFOS	−0.14(-0.15, −0.12)	12.92%	5.01(4.56, 5.55)	5.57(2.65, 11.52)
3/4/5m-PFOS	−0.2(-0.22, −0.19)	18.28%	3.43(3.19, 3.71)	3.83(2.01, 6.1)
2/6m-PFOS	−0.26(-0.28, −0.24)	22.87%	2.67(2.51, 2.85)	2.87(1.67, 4.43)

Note: confident interval (CI), Perfluorooctanoic acid (PFOA), Perfluoropentane sulfonic acid (PFPeS), Perfluorohexane sulfonic acid (PFHxS), Perfluoroheptane sulfonic acid (PFHpS), Linear perfluorooctane sulfonic acid (L-PFOS), Perfluoro-1-methylheptanesulfonate(1m-PFOS), Perfluoro-3/4/5-methylheptanesulfonate(3/4/5m-PFOS), Perfluoro-2/6-methylheptanesulfonate (2/6m-PFOS).

[a] Mean Elimination Rate: From linear mixed model with log-transformed PFAS levels as outcome, adjusted for age, gender, BMI, the coefficient for the variable of time elapsed between the end of distribution of contaminated water and the blood sample collection is the elimination rate.

[b] Mean half-life: Converted from the elimination rate = ln(2)/mean elimination rate.

[c] Median half-life: The individual elimination rates were predicted using the best linear unbiased prediction method, i.e. each participant has an individual predicted elimination rate derived from the model. Then median half-life and percentiles are converted from corresponding individual elimination rates: = ln(2)/ elimination rate.

Table 5
Population average constant elimination rate (yearly) based different rounds of sampling, n = 114.

PFAS	Rounds	Mean Elimination Rate[d] (95% CI)	Mean half-life[e] (95% CI)
PFOA	1—3[a]	−0.33(-0.39, −0.26)	2.1(1.78, 2.67)
	3—6[b]	−0.29(-0.32, −0.26)	2.39(2.17, 2.67)
	6—10[c]	−0.24(-0.27, −0.21)	2.89(2.57, 3.3)
PFPeS	1—3[a]	−0.83(-0.95, −0.71)	0.84(0.73, 0.98)
	3—6[b]	−0.73(-0.79, −0.67)	0.95(0.88, 1.03)
	6—10[c]	−0.72(-0.79, −0.65)	0.96(0.88, 1.07)
PFHxS	1—3[a]	−0.18(-0.22, −0.13)	3.85(3.15, 5.33)
	3—6[b]	−0.16(-0.18, −0.14)	4.33(3.85, 4.95)
	6—10[c]	−0.15(-0.17, −0.13)	4.62(4.08, 5.33)
PFHpS	1—3[a]	−0.19(-0.23, −0.14)	3.65(3.01, 4.95)
	3—6[b]	−0.16(-0.18, −0.14)	4.33(3.85, 4.95)
	6—10[c]	−0.12(-0.15, −0.1)	5.78(4.62, 6.93)
L-PFOS	1—3[a]	−0.28(-0.33, −0.23)	2.48(2.1, 3.01)
	3—6[b]	−0.25(-0.27, −0.23)	2.77(2.57, 3.01)
	6—10[c]	−0.25(-0.27, −0.23)	2.77(2.57, 3.01)
1 m-PFOS	1—3[a]	−0.15(-0.2, −0.11)	4.62(3.47, 6.3)
	3—6[b]	−0.14(-0.16, −0.12)	4.95(4.33, 5.78)
	6—10[c]	−0.14(-0.16, −0.12)	4.95(4.33, 5.78)
3/4/5m-PFOS	1—3[a]	−0.21(-0.26, −0.17)	3.3(2.67, 4.08)
	3—6[b]	−0.2(-0.22, −0.18)	3.47(3.15, 3.85)
	6—10[c]	−0.22(-0.24, −0.19)	3.15(2.89, 3.65)
2/6m-PFOS	1—3[a]	−0.27(-0.32, −0.22)	2.57(2.17, 3.15)
	3—6[b]	−0.25(-0.27, −0.23)	2.77(2.57, 3.01)
	6—10[c]	−0.28(-0.3, −0.26)	2.48(2.31, 2.67)

Note: Perfluorooctanoic acid (PFOA), Perfluoropentane sulfonic acid (PFPeS), Perfluorohexane sulfonic acid (PFHxS), Perfluoroheptane sulfonic acid (PFHpS), Linear perfluorooctane sulfonic acid (L-PFOS), Perfluoro-1-methylheptane sulfonate (1 m-PFOS), Perfluoro-3/4/5-methylheptanesulfonate (3/4/5m-PFOS), Perfluoro-2/6-methylheptanesulfonate (2/6m-PFOS).

[a] 1–3 = between six months and one year after end of exposure.

[b] 3–6 = between one year and two and half years after end of exposure.

[c] 6–10 = between two and half years and four and half years after end of exposure.

[d] Mean Elimination Rate: From linear mixed model with log-transformed PFAS levels as outcome, adjusted for age, gender, BMI, the coefficient for the variable of time elapsed between the end of distribution of contaminated water and the blood sample collection are the elimination rate.

[e] Mean half-life: Converted from the elimination rate = ln(2)/mean elimination rate.

mg/kg had 6–25% shorter half-lives than those with calprotectin < 50 mg/kg. By contrast, increased zonulin levels were mostly associated with slightly longer half-lives (Fig. 7, Table S13), with only one difference reaching significance, for PFHxS ($p = 0.04$).

Based on the partial R^2, age contributed the most to variation of the individual half-lives for all PFAS compounds expect PFPeS (Table S14).

4. Discussion

We estimated the half-lives of eight PFAS in 114 participants, recruited six months after long-term drinking water exposure stopped, who were followed during a four-year period with up to 10 repeated serum measures. The estimated half-lives varied from 0.94 to 5.01 years for different compounds. The inter-individual variations in half-lives were substantial for the different PFAS, with 3 to 5 fold differences between 5th and 95th percentiles. In general, young age, female fertile age, normal estimated glomerular filtration rate, and possibly gut inflammation and lower gut permeability were associated with shorter half-life. There was an evident trend that longer half-life was associated with longer time elapse since end of drinking water exposure. For PFHxS and PFHpS, a longer half-life was associated with higher initial PFHxS levels.

The average half-life for PFHxS was 4.52 years in the current study, with a threefold difference between the 5th and 95th percentiles. To our knowledge, there are the only two other relatively small studies previously reporting half-life for PFHxS after end of exposure. Olsen (2007) reported an estimated half-life of PFHxS of geometric mean 7.3 years based on 24 retired fluorocarbon workers with five-year follow up. This is consistent with our findings as their study population aged 55 to 75 and the half-life in the older 50 + subgroup in our study was 7.25 years. In contrast, Xu et al. (2020a) reported a mean half-life for PFHxS of 2.84 years (95% CI 2.03, 4.43) from 17 airport workers, observed from 2 weeks after end of exposure to 5 months later. Similarly, in the present study we found a shorter half-life in the first period after end of exposure, 6–12 months (3.9 years) compared to the later stage, so still findings fairly consistent between studies.

Our estimated half-life for PFOA with 2.47 years is in the range of values reported in six other studies, with estimates ranging from 1.48 to 3.8 years (Bartell et al., 2010; Brede et al., 2010; Olsen et al., 2007; Worley et al., 2017; Xu et al., 2021). The half-life reported by Bartell 2.3 years for the first year after end of exposure is close to our finding for the first reporting rounds. The estimate by Xu et al. was even shorter, 1.48 years for the first 6 months after end of exposure. The longer half-life values are generally found in cohorts with a longer follow-up, reflecting a possible time-dependent effect on excretion. The age differences across the cohorts, varied background exposure levels and adjustment methods may also contribute to the variation in estimated half-lives.

The half-life of L-PFOS was 2.73 years, whereas for branched isomer 1 m-PFOS, 3/4/5m-PFOS and 2/6m-PFOS were 5.01, 3.43 and 2.67 years, respectively. 1 m-PFOS had the highest half-live estimate, a finding consistent with result from a study in fish (Chen et al., 2015) and a human study with paired urine and serum sample (Zhang et al., 2013). However, our data do not corroborate previous finding of a higher excretion of branched PFOS, with exception of 1 m-PFOS, which previously has been reported (Xu et al., 2021; Zhang et al., 2013).

The estimated half-life for PFOA, PFHxS and linear PFOS were 2.47, 4.52 and 2.73 years. These results are slightly different from the reported value based on two-year follow-up in the same population (Li et al., 2018) due to three reasons. Firstly, in the previous publication, serum PFAS were analysed after each sampling round in order to provide rapid feedback to each participant. However, this may have allowed some batch variation between laboratory analyses. In this paper, all the serum sample in all the ten rounds were analysed in the same batch, thus reducing laboratory variation. Secondly, branched and linear PFOS were identified separately in current study but not in the two-year follow-up. Thirdly, the half-lives presented in the current study are based on serum PFAS levels after subtracting background levels, and the background exposure were not considered in previous publication. Finally, we now have ten rounds providing more data and half-lives could be derived for PFPeS and PFHpS in addition. For these reasons the estimates here are considered more certain than those in the earlier paper.

Our estimated half-life for PFPeS was a mean of 0.94 years (95% CI 0.86–1.02) was slightly longer than the estimate from a smaller cohort with 17 airport workers with an average of 0.63 years (95% CI 0.46 to 1.01 years). For PFHpS with a seven carbon chain length, the estimated half-life from our study (4.55 years, 95 %CI 4.14–5.06) was also much longer than from 17 airport workers (1.35 95% CI 0.77–5.71) (Xu et al., 2021), again indicating time-dependent elimination. However, the confidence intervals of the two studies overlapped.

Differences in elimination with elapsed time after end of exposure for most PFAS were observed, with a greater elimination at the start of the study period for all except the branched PFOS. Time-dependent

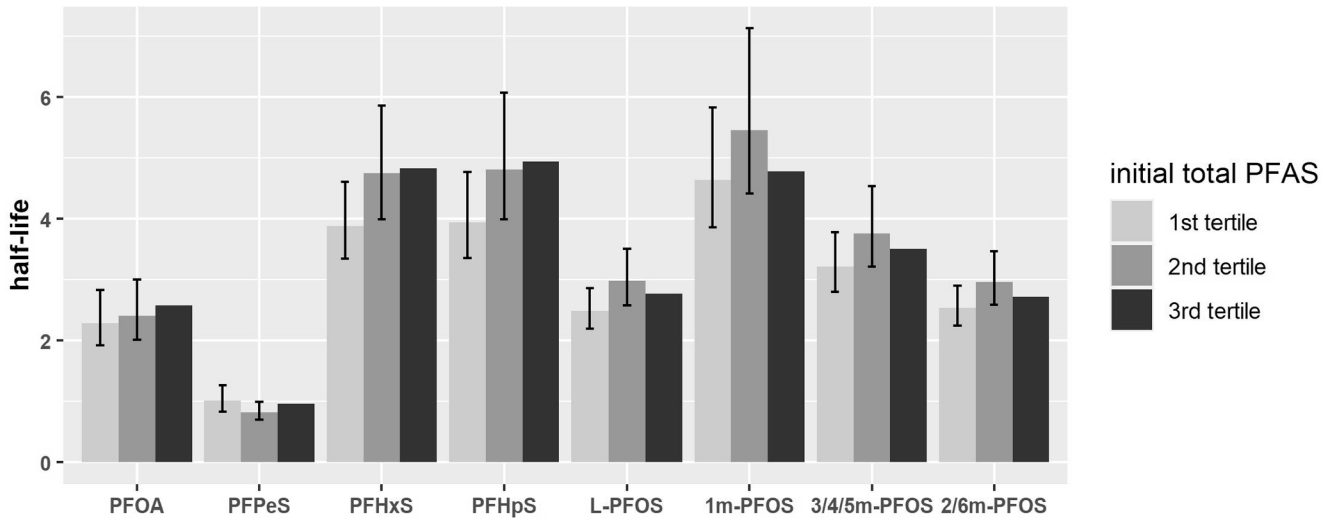

Fig. 1. The mean of estimated half-lives in low-medium–high initial total PFAS levels groups based on 114 participants. The confidence intervals are shown in error bars. The corresponding numeric data of this figure are available in the table S4 Supplemental Material. Note: Perfluorooctanoic acid (PFOA), Perfluoropentane sulfonic acid (PFPeS), Perfluorohexane sulfonic acid (PFHxS), Perfluoroheptane sulfonic acid (PFHpS), Linear perfluorooctane sulfonic acid (L-PFOS), Perfluoro-1-methylheptanesulfonate (1 m-PFOS), Perfluoro-3/4/5-methylheptanesulfonate (3/4/5m-PFOS), Perfluoro-2/6-methylheptanesulfonate (2/6m-PFOS).

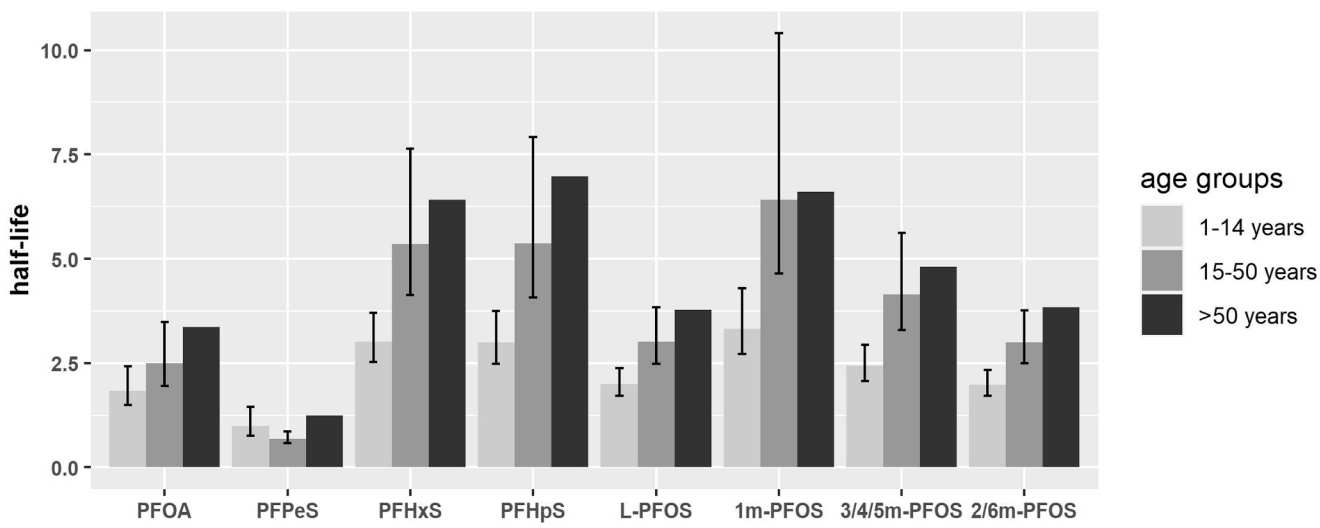

Fig. 2. The mean of estimated half-lives in age groups 1–14 years, 15–50 years and > 50 years based on 114 participants The confidence intervals are shown in error bars. The corresponding numeric data of this figure are available in the table S5 Supplemental Material. Note: Perfluorooctanoic acid (PFOA), Perfluoropentane sulfonic acid (PFPeS), Perfluorohexane sulfonic acid (PFHxS), Perfluoroheptane sulfonic acid (PFHpS), Linear perfluorooctane sulfonic acid (L-PFOS), Perfluoro-1-methylheptanesulfonate (1 m-PFOS), Perfluoro-3/4/5-methylheptanesulfonate (3/4/5m-PFOS), Perfluoro-2/6-methylheptanesulfonate (2/6m-PFOS).

elimination had also been suggested from the C8 study of PFOA (Seals et al., 2011). In the airport workers cohort in Sweden with sampling almost immediate follow-up after end of external exposure, between two weeks to five months, a much shorter half-life of PFOA of 1.48 years was observed (Xu et al., 2020a). This pattern of time-dependent PFAS elimination process with more rapid elimination earlier after the end of exposure could be explained by a multi-compartments mechanism of PFAS elimination, with the serum compartment being replenished from other compartments (i.e. kidneys and liver) at a slow rate.

In a previous human study with C8 data for PFOA (Seals et al., 2011), elimination rates appeared higher for the sub-group with higher prior exposure, but the estimate in this study was limited by ongoing low exposure. From animal studies of rat and monkeys, the differences observed in elimination with increasing dose are indicative that a capacity-limited, saturable renal resorption process must be involved in PFOA and PFOS (Loccisano et al., 2011; Andersen et al., 2006). However, these animal studies involved much higher concentrations than observed in the exposed communities. In our study, indications of increased elimination at lower serum levels of PFAS were observed for most PFAS. We interpret this as likely due to reverse causation, i.e. for a given level of intake those with higher elimination rates would as a consequence already have lower serum levels of PFAS at the start of the observation period.

The inter-individual variations of half-life were substantial, with three to five fold differences across the range from 5th to 95th percentiles. Inter-individual variation in PFAS elimination rates have been previously observed in other populations after end of exposure (Bartell et al., 2010; Brede et al., 2010; Xu et al., 2020a) and in retired fluorochemical production workers (Olsen et al., 2007).

There is a strong effect of age on serum half-life for PFAS. It was expected that the half-life of preteens (≤14 years of age) would be lower than adults', since the PFAS-concentration is greatly reduced as their ongoing growth increases the volume of distribution (Vd) and thus dilutes the PFAS already present in their bodies. However, the Vd for PFAS

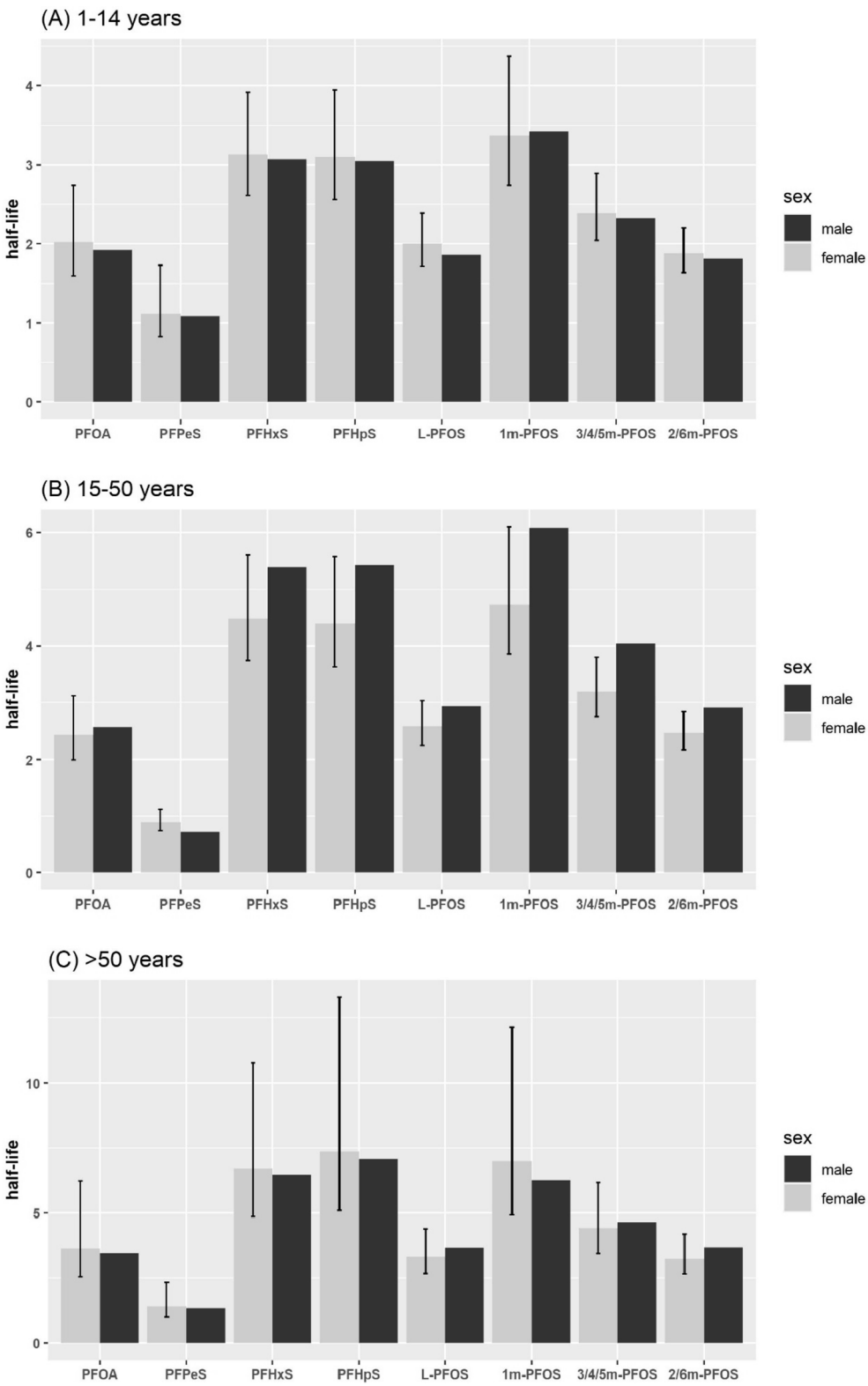

Fig. 3. The mean of estimated half-lives by sex in age strata 1–14 years (n = 29), 15–50 years (n = 52) and > 50 years (n = 33). The confidence intervals are shown as error bars. The corresponding numeric data of this figure are available in the tables S6, S7 and S8 Supplemental Material. Note: Perfluorooctanoic acid (PFOA), Perfluoropentane sulfonic acid (PFPeS), Perfluorohexane sulfonic acid (PFHxS), Perfluoroheptane sulfonic acid (PFHpS), Linear perfluorooctane sulfonic acid (L-PFOS), Perfluoro-1-methylheptanesulfonate (1 m-PFOS), Perfluoro-3/4/5-methylheptane sulfonate (3/4/5m-PFOS), Perfluoro-2/6-methylheptanesulfonate (2/6m-PFOS).

should not increase or decrease for adults, since PFAS primarily is distributed in serum, kidney and liver tissue (Loccisano et al., 2011), which are not prone to major changes of Vd in adulthood. The trend for elimination to decline with age continues at older age, suggesting that in addition to growth-induced dilution, elimination of the substances depends on other age-related factors. Notably, differences in renal function or gut excretion are unlikely the explanations for the age effect, since the age effect persisted after adjustment for eGFR, the urine/serum ratio and markers of gut inflammation and reduced permeability. Thus, there are likely other age-associated factors to be explore.

Females in fertile ages had a shorter half-life for all analysed compounds except PFPeS. The longer PFPeS half-life for women unlike other PFAS may be a chance finding or it might be explained if there were some ongoing but unmeasured exposure in women more than men, for

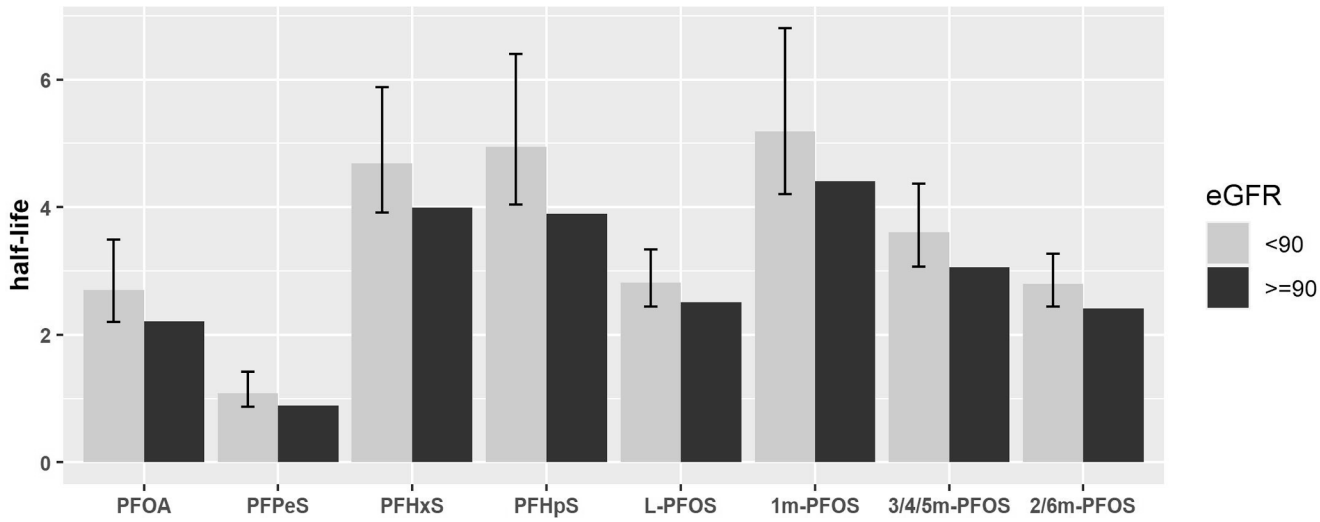

Fig. 4. The mean of estimated half-lives in those eGFR below or above 90 based 109 participants. The confidence intervals are shown aserror bars. The corresponding numeric data of this figure are available in the table S10 Supplemental Material. Note: Perfluorooctanoic acid (PFOA), Perfluoropentane sulfonic acid (PFPeS), Perfluorohexane sulfonic acid (PFHxS), Perfluoroheptane sulfonic acid (PFHpS), Linear perfluorooctane sulfonic acid (L-PFOS), Perfluoro-1-methylheptanesulfonate (1 m-PFOS), Perfluoro-3/4/5-methylheptanesulfonate (3/4/5m-PFOS), Perfluoro-2/6-methylheptanesulfonate (2/6m-PFOS).

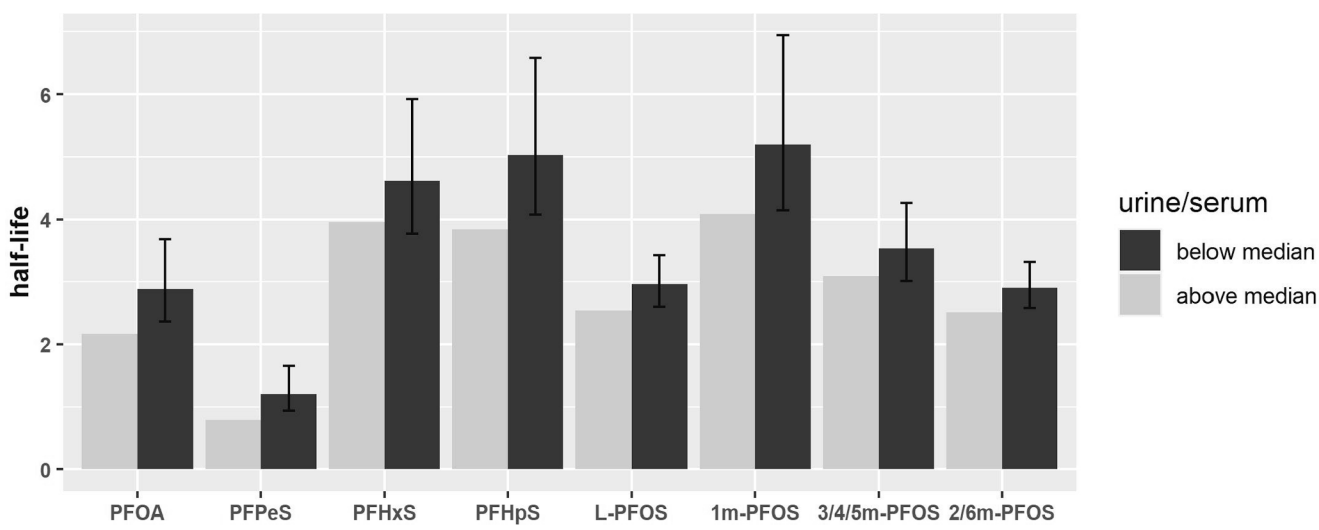

Fig. 5. The mean of estimated half-lives in high urine/serum ratio and low groups (n = 62). The confidence intervals are shown as error bars. The corresponding numeric data of this figure are available in the table S11 Supplemental Material. Note: Perfluorooctanoic acid (PFOA), Perfluoropentane sulfonic acid (PFPeS), Perfluorohexane sulfonic acid (PFHxS), Perfluoroheptane sulfonic acid (PFHpS), Linear perfluorooctane sulfonic acid (L-PFOS), Perfluoro-1-methylheptanesulfonate (1 m-PFOS), Perfluoro-3/4/5-methylheptanesulfonate (3/4/5m-PFOS), Perfluoro-2/6-methylheptanesulfonate (2/6m-PFOS).

example PFPeS has been reported in some lipsticks (Whitehead et al., 2021). Sex differences in renal excretion have been reported in animals (Kudo et al., 2002), but not in human subjects (Brede et al., 2010; Harada et al., 2005). There were no statistically significant sex differences in the younger and older age groups. Therefore the differences in elimination in the fertile ages are likely due to maternal elimination route, i.e. losses during menstruation, childbirth and breastfeeding (Wong et al., 2014), but may also be influenced by other factors such as a possible estrogenic induction of tubular secretion, as has been found in rats (Han et al., 2012). We observed that higher eGFR was associated with shorter half-lives for all the PFAS. Pharmacokinetic studies has demonstrated that PFOA and PFOS are distributed primarily in serum, kidney and liver and are substantially excreted by the kidneys (Loccisano et al., 2011). In humans, PFOA is both freely filtered in the glomeruli and actively secreted from the blood into urine through the proximal tubule cells. PFOA is then almost completely reabsorbed back into the blood (Loccisano et al., 2011). Since the renal excretion is partly dependent on glomerular filtration, it is understandable that a higher eGFR contributes to a higher amount of urinary excretion and shorter half-life. Similarly, it is understandable that a higher urine/serum ratio is associated with a higher elimination rate, since a higher ratio means that a higher fraction of PFAS is excreted through the urine. Increased PFAS levels were associated with decreased urine to serum ratio reflecting a capacity-limited of urinary elimination. However, the fact that urine/serum ratio is poorly correlated with eGFR strongly suggests that there are other mechanisms determining the urine/serum ratio. One such mechanism could be tubular reabsorption and secretion, which could differ between individuals, for example through genetic or epigenetic differences.

PFAS can pass into the gut and be excreted, though most is reabsorbed (Fragki et al., 2021). Variation in the efficiency of reabsorption would thus be reflected in effects on elimination rates. We observed the shorter half-lives of PFAS, especially for PFOA, are associated with higher calprotectin in faeces, a biomarker that is routinely used as an

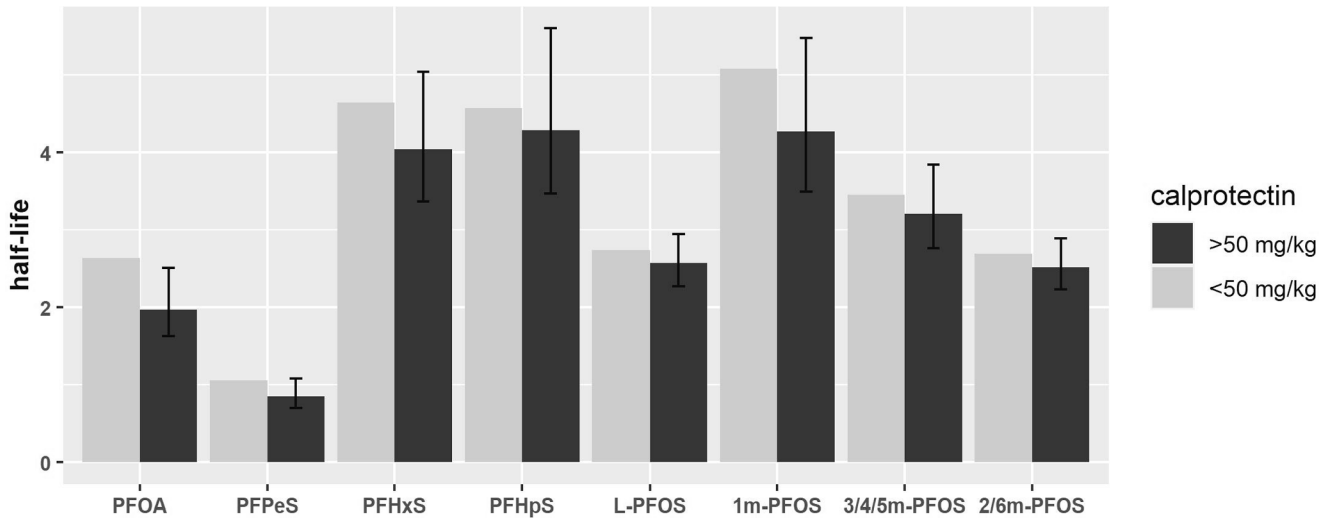

Fig. 6. The mean of estimated half-lives in those calprotectin below or above 50 (n = 57). The confidence intervals are shown as error bars. The corresponding numeric data of this figure are available in the table S12 Supplemental Material. Note: Perfluorooctanoic acid (PFOA), Perfluoropentane sulfonic acid (PFPeS), Perfluorohexane sulfonic acid (PFHxS), Perfluoroheptane sulfonic acid (PFHpS), Linear perfluorooctane sulfonic acid (L-PFOS), Perfluoro-1-methylheptanesulfonate (1 m-PFOS), Perfluoro-3/4/5-methylheptanesulfonate (3/4/5m-PFOS), Perfluoro-2/6-methylheptanesulfonate (2/6m-PFOS).

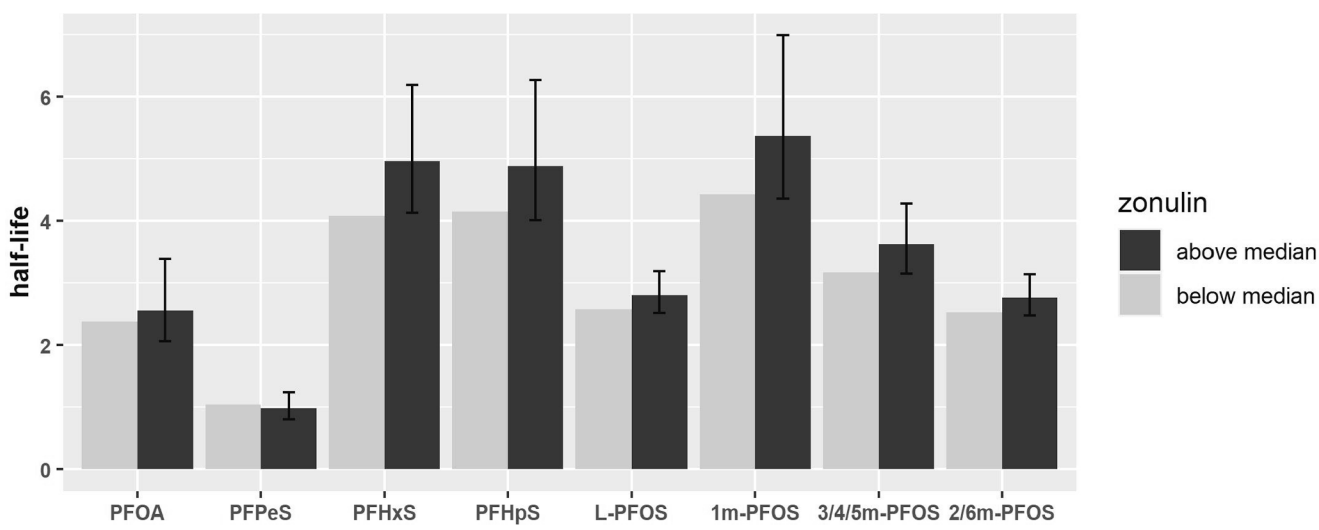

Fig. 7. The mean of estimated half-lives in those zonulin below or above median (n = 57). The confidence intervals are shown as error bars. The corresponding numeric data of this figure are available in the table S13 Supplemental Material. Note: Perfluorooctanoic acid (PFOA), Perfluoropentane sulfonic acid (PFPeS), Perfluorohexane sulfonic acid (PFHxS), Perfluoroheptane sulfonic acid (PFHpS), Linear perfluorooctane sulfonic acid (L-PFOS), Perfluoro-1-methylheptanesulfonate (1 m-PFOS), Perfluoro-3/4/5-methylheptanesulfonate (3/4/5m-PFOS), Perfluoro-2/6-methylheptanesulfonate (2/6m-PFOS).

indicator of inflammation in the gastrointestinal tract (Erbayrak et al., 2009). This association suggests that intestinal inflammation is related to reduced reabsorption, increasing the fecal PFAS-elimination. In contrast, though with weaker associations, a longer half-life was associated with higher levels of zonulin. Zonulin has been suggested as a marker of intestinal permeability (Malíčková et al., 2017), as it has been found to be associated with increased intestinal permeability in in vitro studies (Fasano, 2012; Sapone et al., 2006). However, zonulin is not commonly used for in vivo studies or as a clinical marker, so its use in understanding these results is limited. Further studies are needed to understand the mechanisms of PFAS transport in the gut and their effects on PFAS elimination.

5. Conclusions

The present study gives half-life estimates of eight PFAS, especially including the branched PFOS-isomers. We found that the most important determinants of a shorter half-life were young age and better kidney function, followed by female sex during their fertile period. Markers of gut inflammation and reduced permeability were also associated with shorter half-life, but should be investigated further. These results provide observational support for elimination routes – renal, fecal and maternal.

The result also suggests a time-dependent PFAS elimination process, with more rapid elimination in the first year after the end of exposure, than later.

CRediT authorship contribution statement

Ying Li: Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing. **Axel Andersson:** Writing – original draft, Writing – review & editing. **Yiyi Xu:** Methodology, Writing – review & editing. **Daniela Pineda:** Investigation, Writing – review & editing. **Carina A Nilsson:** Investigation, Writing – review & editing.

Christian H. Lindh: Resources, Investigation, Writing – review & editing. **Kristina Jakobsson:** Conceptualization, Writing – review & editing, Supervision, Funding acquisition. **Tony Fletcher:** Conceptualization, Writing – review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envint.2022.107198.

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